Red-Blue Hamiltonian Tomography via the Operator Lightcone Wavemap

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Abstract

We introduce a model-free method for Hamiltonian tomography based on the operator spreading wavefront. Given two tensor network simulations of the same quantum lattice — one with the nominal Hamiltonian and one with a single perturbed bond — the signed cross-entropy of the non-identity Pauli weight field $\mathcal{L}(v,t) = -\ln(n_{\text{pert}}/n_{\text{base}})$ identifies the perturbed bond by its causal signature: a red-blue sign flip at the two endpoints, visible from the first wavefront crossing. Combining measurements from p strategically placed observation centers via an ensemble log-likelihood sum, the broken bond is identified with an 82-fold margin over the next candidate on a 7×7 lattice, even though individual centers give only 2/6 correct answers. The method requires no model assumptions, no device calibration access, and maps directly to hardware via zero-noise extrapolation and local Pauli measurements.

1 Introduction

Identifying miscalibrated or anomalous couplings in a quantum processor is a central challenge in quantum hardware characterization. Standard approaches require explicit Hamiltonian learning — fitting a parameterized noise or coupling model to measurement data — which scales poorly with system size and depends on model assumptions.

We propose a different approach: measure the operator spreading wavefront $n(v,t)$, the non-identity Pauli weight at site v and cycle t, from multiple initialization sites. A perturbed bond creates a causal signature in the wavefront — a sign flip in the cross-entropy field between the two endpoints — that propagates no further. The broken bond is exactly where the wavefront sign changes.

The method is inspired by seismic tomography: place sensors (observation centers) at strategic positions, emit a wave (initialize a local operator), and read the arrival pattern. The perturbed bond is the anomaly in the medium.

The red-blue color scheme is a deliberate reference to the red-blue pebble game of Hong and Kung [**?**], which characterizes I/O complexity as a bottleneck in computational graphs. Here the roles are analogous: red sites (upstream, wave blocked) and blue sites (downstream, wave starved) identify the bond that limits operator propagation, exactly as red and blue pebbles identify the I/O bottleneck in a computation.

2 Setup

Observable. We simulate the kicked isotropic Heisenberg model on a 7×7 rectangular lattice (49 sites, 84 bonds) using a GPU-accelerated belief propagation tensor network simulator with Trotter

step $\varepsilon = 0.10$, bond dimension $\chi = 50$, and truncation cutoff $\delta = 10^{-3}$.

The non-identity Pauli weight at site v and cycle t is:

$$n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2 \tag{1}$$

where $O_c(t) = \sum_P c_P(t) P$ is the evolved operator initialized at center c. By normalization $\sum_P c_P^2 = 1$, $n(v,t) \in [0,1]$ is a probability.

Perturbation. C_breaker selects one bond b^* and replaces its Pauli transfer matrix (PTM) with the identity, blocking all operator propagation across that bond. This is the maximally disruptive perturbation — a perfect switch.

p-center placement. We place $p = 6$ observation centers using a greedy geodesic bond cover: the minimum set of sites such that every bond is within $d_{\max} = 2$ hops of at least one center. For the 7×7 lattice: centers at sites $\{6, 10, 25, 30, 32, 34\}$, covering all 84 bonds.

3 The Cross-Entropy Wavemap Observable

Signed cross-entropy. For each center k, run two simulations (baseline and perturbed) and compute the signed cross-entropy at first baseline arrival:

$$\mathcal{L}_k(v) = -\ln \frac{n^k_{\text{pert}}(v,\, t^k_{\text{first}}(v))}{n^k_{\text{base}}(v,\, t^k_{\text{first}}(v))} \tag{2}$$

where $t^k_{\text{first}}(v) = \min\{t : n^k_{\text{base}}(v,t) > \varepsilon_{\text{th}}\}$ is the first cycle at which the baseline wave reaches site v from center k.

At $t^k_{\text{first}}(v)$, the baseline wave has just arrived while the perturbed wave may be blocked or delayed — this is the moment of maximum signal and minimum chi-saturation noise.

Sign interpretation.

- $\mathcal{L}_k(v) > 0$: perturbed weaker (red) — upstream of broken bond; wave cannot cross
- $\mathcal{L}_k(v) < 0$: perturbed stronger (blue) — downstream shadow; wave never arrives
- $\mathcal{L}_k(v) = 0$: no perturbation

The broken bond is exactly where the sign changes — between the red and blue endpoints (Figure **??**).

Ensemble log-likelihood. Sum the signed cross-entropy fields across all p centers:

$$\mathcal{L}_{\text{ens}}(v) = \sum_{k=1}^{p} \mathcal{L}_k(v) \tag{3}$$

This is a log-likelihood ensemble: centers on the same side of the broken bond reinforce each other; centers on opposite sides partially cancel noise. Silent centers (wave never reaches) contribute zero and do not corrupt the sum.

4 Bond Detection

Bond score. For each bond b connecting sites v_a and v_b:

$$S(b) = \mathcal{L}_{\text{ens}}(v_a) \times \mathcal{L}_{\text{ens}}(v_b) \tag{4}$$

A negative product indicates a sign flip across the bond — the causal signature of the broken bond. Rank bonds by most negative $S(b)$.

Results. Table 1 shows the individual and ensemble results for bond 12 (sites 15 $leftrightarrow$16, identity PTM perturbation, 7×7 lattice, $\chi = 50$, 5 cycles).

Center	Side	$\mathcal{L}(15)$	$\mathcal{L}(16)$	Result
10	upstream	$+0.87$	-0.02	✓CORRECT
30	upstream	$+0.37$	-0.04	✓CORRECT
25	downstream	$+0.03$	-0.04	× WRONG
32	downstream	$+0.00$	-0.03	× WRONG
6	too far	0	0	silent
34	too far	0	0	silent
Ensemble	—	$+1.23$	-0.13	✓CORRECT

Table 1: Signed cross-entropy at bond 12 endpoints (sites 15,16). Ensemble score: $S(12) = -0.164$, next bond $= -0.002$ (82× margin).

The ensemble identifies bond 12 with an 82-fold margin over the next candidate, despite only 2 of 6 individual centers giving the correct answer. The wrong individual answers — centers 25 and 32 — both correctly identify site 16 as a bond endpoint; their error is only in declaring the wrong adjacent bond. The ensemble product cancels these partial errors.

5 The Seismic Approach

Running all p centers simultaneously (seismic initialization $O(0) = \sum_k Z_{c_k}$) produces the same ensemble field in a single simulation. At each site v, the first-arrival signal is dominated by the nearest center — automatically selecting the strongest directional component. The sign flip at the broken bond endpoints emerges without any explicit path tracing or center-by-center analysis.

The seismic approach reduces $2p$ simulations (p baselines + p perturbed) to 2 simulations at the cost of mixing center contributions. For the purpose of bond detection the mixing is benign: the sign flip at the broken bond is preserved since the nearest center to each endpoint dominates at t_{first}.

6 Discussion

Connection to hardware. The observable $n(v, t)$ is accessible on real quantum hardware via local Pauli measurements after ZNE gate-folding circuits [1]. The cross-entropy $\mathcal{L}(v)$ requires only the ratio of two measurements — no absolute calibration. The p-center placement is determined by the graph topology alone, independent of the Hamiltonian.

Causal structure. The method exploits the Lieb-Robinson bound: information cannot propagate faster than v_{hop} hops/cycle. A broken bond creates a causal anomaly — the wavefront on one side cannot inform the other. This is detectable without knowing v_{hop} explicitly; the first-arrival time $t_{\text{first}}(v)$ is observed directly.

Limitations. The current experiment uses the identity PTM — a maximally disruptive perturbation. Realistic coupling deviations ($\delta J \sim 10\%$) will produce smaller signals. The sensitivity threshold and the extension to multiple simultaneous bond perturbations are open questions.

7 Conclusion

We have demonstrated that the operator lightcone wavemap enables model-free Hamiltonian tomography via a simple signed cross-entropy observable. The broken bond is identified by its causal signature — a sign flip in $\mathcal{L}(v)$ at its two endpoints — with an 82-fold margin using an ensemble of 6 observation centers on a 7×7 lattice. The seismic multi-center initialization achieves the same result in a single simulation. The method requires no model assumptions and connects directly to hardware via ZNE and local Pauli measurements.

References

[1] Tudor Giurgica-Tiron, Yousef Hindy, Ryan LaRose, Andrea Mari, and William J. Zeng. Digital zero noise extrapolation for quantum error mitigation. *2020 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 306–316, 2020.

[2] Jia-Wei Hong and H. T. Kung. I/O complexity: The red-blue pebble game. In *Proceedings of the Thirteenth Annual ACM Symposium on Theory of Computing (STOC '81)*, pages 326–333, 1981.